CS3 Rubric — Movie Poster Visual Features & Box Office Revenue

DS 4002 Case Study

General Description

Using a provided dataset of ~1000 (or 21,000+) films with pre-extracted visual features from their official movie posters, you will investigate whether quantitative poster characteristics, such as color composition, brightness, saturation, and face density, are statistically significant predictors of worldwide theatrical box office revenue. You will produce a short analytical report (3–5 pages, PDF) with reproducible code.

Why Am I Doing This?

Movie studios invest heavily in poster design as a key marketing tool. This case study asks you to apply the data science pipeline, from exploratory analysis through statistical modeling and predictive evaluation, to a creative and real-world question. You will practice working with image-derived features, interpreting regression output, and communicating a data-driven conclusion to a non-technical audience.

Spec Category & Details

Category	Details
Data	Use the provided `sample_poster_features.csv` dataset (~1,000 films) located in the `DATA/` folder. See `DATA_DICTIONARY.md` for variable definitions. If you wish to expand the dataset, follow `DATA_PREPARATION.md`.

Category	Details
Exploratory Data Analysis	Produce a correlation heatmap of all numeric visual features against log-transformed worldwide revenue (`log_worldwide`). Include at least one additional bivariate visualization (e.g., box plot by release window). Report exact correlation coefficients.
Statistical Modeling	Fit a Multiple Linear Regression (OLS) model predicting `log_worldwide` from the visual features and release window controls. Report R-squared, p-values, and interpret which predictors (if any) are statistically significant at = 0.05.
Predictive Evaluation	Train a machine learning model (e.g., Random Forest) using an 80/20 train/test split. Report MAE, RMSE, and R-squared on the test set. Compare its performance to the linear model. Include a feature importance visualization.
Report	3-5 page PDF. Must include: (1) an introduction framing the research question, (2) a methods section describing feature extraction and modeling, (3) results with at least 3 figures/tables, and (4) a conclusion addressing whether visual features predict box office success and what factors might be missing.
Code	All analysis code must be in a Jupyter Notebook (`.ipynb`) or Python script. It should be runnable by the grader with the provided data. Use the `SCRIPTS/` folder. A `starter_analysis.ipynb` is provided to get you started.
Repository	Submit a link to a GitHub repository containing your report (PDF), code, and any additional materials.

How Will I Know I Have Succeeded?

Component	Meets Expectations	Does Not Meet Expectations
EDA	Correlation heatmap with all numeric features. At least one other visualization. Exact coefficients reported.	Missing heatmap, no additional visualization, or coefficients not reported.
OLS Regression	Model is fitted with visual + release window features. R-squared, p-values, and coefficients are reported and interpreted correctly.	Model is missing, uses wrong target variable, or output is not interpreted.
Predictive Model	ML model trained on 80/20 split. MAE, RMSE, and R^2 reported on test set. Feature importance plot included. Performance compared to OLS.	No ML model, or metrics not reported, or no comparison to OLS.
Report Quality	Clear writing. Research question stated. Methods described. Results supported by figures. Conclusion addresses the core question.	Unclear writing, missing sections, figures without interpretation, or no conclusion.
Reproducibility	Code runs without errors on the provided dataset. Notebook or script is well-organized with comments.	Code crashes, relies on unavailable files, or is disorganized.

References

See the `SUPPLEMENTAL_MATERIALS/` folder for: - A blog-style explainer article motivating the project and introducing key concepts - Links to technical documentation for the tools and APIs used

Acknowledgements

This case study is based on a DS 4002 project by Rowan Rosenblum (Spring 2025). The analysis examined 21,000+ films to test whether poster visual features predict global box office revenue. The provided sample dataset is a curated subset for instructional use.